UGANDA ADVANCED CERTIFICATE OF EDUCATION
PRINCIPAL MATHEMATICS PAPER 1
COMPETENCY-BASED ASSESSMENT ITEMS

Time: 3 Hours

INSTRUCTIONS TO CANDIDATES:

- This paper consists of three sections: **A** (Algebra), **B** (Geometry), and **C** (Calculus).
- Answer **four** items in total: **one** item from Section A, the **compulsory** item from Section B, and **two** items from Section C.
- All items carry equal marks. Show all mathematical modeling, structural equations, and final interpretations clearly.

SECTION A: ALGEBRA

Answer exactly ***one*** *item from this section.*

ITEM 1 (Agritech Automation & Environmental Expansion Modeling)

Scenario:
An agritech startup is designing an automated solar-powered irrigation valve system to optimize water flow delivery across smallholder farming blocks in arid regions. The operational flow rate factor must maintain a strict mechanical equilibrium balance equation to prevent pressure drops in the primary pipeline network (Chen et al., 2023). Concurrently, environmental researchers track the long-term spatial distribution of organic compost nutrients spread over these large farming plots. The distribution density of these nutrients across the fields expands systematically, requiring the logistics team to model the behavioral ratios of consecutive high-order variables to forecast seasonal fertilizer supply requirements (Al-Kaisi & Kwaw-Mensah, 2020).

Tasks:

a) By applying a suitable algebraic substitution (e.g., let $u = x^{\frac{2}{3}}$), solve the stability equation $64x^{\frac{2}{3}} + x^{-\frac{2}{3}} = 20$ to determine all mathematically viable real values for the irrigation water flow optimization threshold x.

b) Determine the exact ratio of the coefficient of x^7 to the coefficient of x^8 in the algebraic distribution expansion of $\left(3x + \frac{2}{3}\right)^{17}$ to guide resource distribution planning.

c) Analyze the structural convergence conditions for an infinite canopy cover progression modeled as an infinite geometric series $\left(\frac{x}{2x-1}\right) + \left(\frac{x}{2x-1}\right)^2 + \left(\frac{x}{2x-1}\right)^3 + \cdots = \frac{3}{4}$, and find the exact value of x for which this environmental canopy model stabilizes.

ITEM 2 (Smart Grid Power Management & Eco-System Stability Analytics)

Scenario:
A power systems engineer is creating a micro-grid load balancer to manage renewable energy inputs from community solar arrays in rural Uganda. The functional power loads across two primary distribution lines carry complex mathematical variables governed by two distinct complex numbers, w and z, which must satisfy simultaneous network balance parameters (Mahdavi et al., 2022). In another phase of the project, a software engineer models the long-term ecological impact of the grid infrastructure on local biodiversity. The population density variance of indicator bird species over the operational area is mapped by a higher-order polynomial function where field data indicates that the ecosystem maintains high structural resilience and perfect stability profiles whenever the algebraic factors cleanly divide the polynomial mapping model (Sutherland et al., 2021).

Tasks:

a) Solve the simultaneous complex linear equations to find the distinct properties of variables w and z, presenting your final solutions in standard rectangular form $(x + iy)$:

$$\begin{aligned} 3z + w &= 9 + 11i \\ w - z &= -8 - 2i \end{aligned}$$

b) Utilizing the ecological boundary conditions that $(x-1)$ and $(x+1)$ are exact linear factors of the biodiversity mapping polynomial $f(x) = ax^4 + 7x^3 + x^2 + bx - 3$, calculate the exact values of the design constants a and b.

c) Find the full set of real values of x for which the biodiversity status remains highly secure, satisfying the positive operational condition $f(x) > 0$.

SECTION B: GEOMETRY

*This section is **compulsory**.*

ITEM 3 (Drone-Based Precision Agriculture & Logistics Infrastructure Design)

Scenario:
An agritech drone flight team is mapping out an optimized distribution center for an organic fertilizer marketplace. In a localized 3D coordinate system, the boundaries of the primary drone sorting yard are marked by four ground sensors forming a flat quadrilateral zone in the shape of a parallelogram $ABCD$ (Radoglou-Grammatikis et al., 2020). To protect drone flight vectors from electromagnetic interference, the communications team establishes a horizontal safety plane running between two telecommunication towers A' and B' that must be positioned such that it remains exactly equidistant from both towers at all points. A secondary laser guidance beam runs through the zone along a continuous vector line path, which must pass safely through the main facility firewall structural shield plane (Vrochidou et al., 2021).

Tasks:

a) Utilizing vector cross-products ($\vec{AB} \times \vec{AD}$), calculate the precise surface area of the primary drone sorting yard enclosed within the parallelogram vertices $A(1, 3, 2)$, $B(2, -1, 1)$, $C(-1, 2, 3)$, and $D(-2, 6, 4)$.

b) Formulate the exact Cartesian equation of the communications safety plane that is equidistant from the two towers $A'(1, 3, 5)$ and $B'(2, -4, 4)$.

c) Find the precise spatial coordinates of the point P where the laser guidance beam $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z+4}{2}$ intersects the main facility firewall structural shield plane represented by the coordinate equation $4x + 5y + 6z = 87$, and calculate the exact angle of inclination between the beam path and the plane to verify flight line calibrations.

SECTION C: CALCULUS

*Answer exactly **two** items from this section.*

ITEM 4 (Biomass Energy Conversion Dynamics & Geolocation Error Correction)

Scenario:
A biochemical analyst is measuring gas production rates inside an anaerobic organic waste digester tank. The internal biogas production rate G varies dynamically as a function of operational cycle time t according to a periodic trigonometric model (Kyazze et al., 2021). To maintain safe tank conditions, an automated sensor network estimates system parameters using localized power series expansions. The primary calibration tracking curve for the sensor's microprocessor is represented by an inverse trigonometric function where a digital tracking offset at an angle close to standard parameters can be estimated using small increments (Aghbashlo et al., 2022).

Tasks:

a) Apply Maclaurin's theorem to expand the microprocessor calibration curve $(x+1)\sin^{-1}x$ into a polynomial series up to the term in x^2 to facilitate fast real-time computational processing.

b) Determine the exact initial time $t > 0$ when the digester tank reaches its absolute peak biogas production rate given by the function $G(t) = 8000[1 - \sin(2\pi t - 3)]$, and calculate the maximum gas production volume sustained at that instant.

c) Using the calculus of small changes ($\delta y \approx \frac{dy}{dx}\delta x$), show that the geographic alignment offset for a solar tracker tilted near a standard reference angle can be modeled analytically as:

$$\cos 44.6^\circ = \frac{\sqrt{2}}{2}\left(\frac{900 + 2\pi}{900}\right)$$

ITEM 5 (Sustainable Architectural Design & Irrigation Hydrodynamics)

Scenario:
A sustainable architect is designing the structural profile for a rainwater harvesting canopy wall. In the design blueprints, the curved edge of the structural canopy is modeled accurately by a quadratic relationship representing a conic section curve supported by a linear reinforcement beam constructed to lie exactly parallel to the directrix of the curve (Amos et al., 2024). In a separate sub-project, irrigation engineers are building an optimal water-blending basin. The total volume capacity of the mixing unit is generated by rotating the two-dimensional region bounded between two overlapping structural curves through one full revolution around the horizontal x-axis (Burt & Styles, 2019).

Tasks:

a) By completing the square on the quadratic boundary equation $3y^2 - 18y + 2x + 37 = 0$, express it in standard vertex form. Prove mathematically that the canopy profile is a parabola, and determine the exact linear equation of the reinforcement beam (the directrix).

b) Set up a definite integral for the solid of revolution, and show that the total volume capacity generated by rotating the region bounded by $y^2 = 32x$ and $y = x^3$ around the x-axis is exactly $\frac{320\pi}{7}$ cubic units.

c) Evaluate the definite integral $\int_0^{\frac{\sqrt{3}}{2}} \frac{1}{(1-x^2)^{\frac{3}{2}}}\, dx$ using a suitable inverse trigonometric substitution to calculate the material structural loading constant.

ITEM 6 (Hydrological Reservoir Management & Smart Fluid Mechanics)

Scenario:
An environmental engineering team is managing a community water purification reservoir tank in a semi-arid district. A minor structural leakage occurs at the base of the tank, causing treated water to escape onto the surrounding water table where hydrological data confirms that the rate of reduction of the water mass is directly proportional to the instantaneous mass M of water remaining inside the reservoir tank at any elapsed time t (Adoga et al., 2023). To optimize the system's performance, the engineering lab also analyzes the structural flow throughput behavior of a separate multi-stage filtration system modeled by a rational fraction expression (Crittenden et al., 2022).

Tasks:

a) Formulate a first-order differential equation modeling the relationship between the remaining water mass M, time t, and the decay constant k. Solve this differential equation to prove it reduces to the exponential model $M = M_0 e^{-kt}$, where M_0 represents the initial mass of water.

b) The emergency repair crew takes exactly 2 hours to arrive at the reservoir site. Given that after 1 hour of the leak, precisely 10% of the initial water mass has been lost, calculate the exact percentage of the water mass remaining in the reservoir tank when the crew arrives at the 2-hour mark.

c) Resolve the filtration flow throughput expression $\frac{x^3+9x^2+28x+28}{(x+3)^2}$ into its constituent partial fractions. Hence, prove that the cumulative flow over the initial filtration cycle satisfies:

$$\int_0^1 \frac{x^3 + 9x^2 + 28x + 28}{(x+3)^2}\, dx = \frac{1}{3}\left(10 + \ln\frac{4}{3}\right)$$